Neha Nepal

nepalneha25@gmail.com · (650) 382-8343 · Sunnyvale, CA · www.linkedin.com/in/nehanepal · https://nehanepal.github.io/

EDUCATION

University of California, Santa Barbara, Santa Barbara, CA *June 2026*
Bachelor of Science (B.S.), Mechanical Engineering
Relevant Coursework: COMSOL, Aerodynamics, Mechatronics, Heat Transfer, Fluid Mechanics, Wind and Tidal Energy, Materials

PROJECT EXPERIENCE

Team Lead *September 2025 - June 2026*
Northrop Grumman - Sponsored Capstone, UCSB
- Led a five-person team through **design** and **prototyping** of a specialized camera housing to ensure hardware survivability.
- Designed a camera housing to be capable of protecting imaging hardware across temperatures from -180°C to 130°C under **vacuum** conditions while maintaining operational conditions for internal components.
- Validated thermal and structural performance using **COMSOL** and **Solidworks** simulations.
- Integrated a thermal management system using **valves**, **heaters** and **chillers** to autonomously regulate internal temperatures.
- Performed **nitrogen** and **vacuum** testing to validate performance under representative aerospace operating conditions.

Robotics Design Engineer *March - June 2025*
Window Washer, UCSB
- Designed an autonomous, **Arduino**-powered window washer under a $150 budget, from sketch to finished assembly.
- Reduced friction by 40% and weight by 15%, improving motor efficiency and battery life through five design iterations.
- Applied **fluid mechanics** to develop a spray system, achieving 95% dirt removal within five passes during testing.
- Demonstrated prototype performance to faculty and industry professionals through live system testing.

Design Engineer *March - June 2025*
Airfoil Optimization, UCSB
- Designed a 50 m turbine blade using the S8037 airfoil, optimized for a 12 m/s wind speed and a Tip Speed Ratio of 6.
- Achieved 1.03 MW output and a coefficient of performance of Cp = 0.5 using **MATLAB** and **Blade Element Momentum**.
- Performed **Solidworks** deflection analysis across wind speeds of 4–20 m/s, verifying blade deformation was within allowable limits
- Modeled wake interactions in MATLAB using the **Entrainment** model to maximize wind farm power output while minimizing turbine interference.

Design Engineer *January - March 2023*
Air Motor, UCSB
- Machined and assembled an air motor with aluminum using **CNC mills**, **lathes** and **precision hand tools**.
- Improved motor efficiency by refining machining tolerances and surface finishes, reducing friction.
- Exceeded performance targets by 25%, achieving a tachometer verified rotational speed of 3,000 RPM.

PROFESSIONAL EXPERIENCE

Project Engineering Intern *June - September 2025*
Western Allied Mechanical, Union City, CA
- Audited mechanical drawings across 10 HVAC projects, identifying discrepancies that saved 10+ project manager hours.
- Conducted field inspections for $2-13M healthcare and laboratory projects, verifying installations met design specifications.
- Prevented costly rework by identifying installation discrepancies and coordinating resolutions between field crews and Project Managers.
- Streamlined **RFI** and **submittals** with real-time procurement logs, reducing lead-time bottlenecks for critical equipment.

Operations Assistant *June 2024 - June 2025*
UCSB Recreation, Santa Barbara, CA
- Managed front desk operations during peak hours, training over 30 new hires to maintain service standards and efficiency.
- Maintained **CPR** certification to support safe operations and emergency response in a high-traffic recreation facility.

SKILLS

CAD & Design: Solidworks, Fusion 360, GD&T, Design for Manufacturing (DFM)
Simulation: COMSOL, Finite Element Analysis (FEA), Thermal Analysis, Structural Analysis
Analysis & Programming: MATLAB, Python, Arduino IDE
Manufacturing: CNC Milling, Lathe, 3D Printing, Rapid Prototyping, Testing & Validation